CHAPTER 5 | Data and Knowledge Management

CHAPTER ROADMAP

Good data architecture and governance turn records into reliable operational and analytical assets.

Learning outcomes

1. Explain and apply: Data organisation.
2. Explain and apply: Relational databases.
3. Explain and apply: Queries and transactions.
4. Explain and apply: Data warehousing and analytics.
5. Explain and apply: Big data and quality.
6. Explain and apply: Knowledge management.

Concept map

BUSINESS QUESTION → DATA / PEOPLE / PROCESS → INFORMATION SYSTEM → ACTION → MEASURABLE OUTCOME

How to use this chapter

Read the six topic pages, explain the definitions aloud, study the scenario, and answer the questions without looking. Then check the model answers. This guide is original educational commentary, not a reproduction of the textbook.

Big picture

Good data architecture and governance turn records into reliable operational and analytical assets. The central skill is to connect terminology to a real organisational decision. Whenever you meet a new concept, identify its purpose, required inputs, expected benefits, limitations, and the people responsible for it.

Data organisation

Core explanation

Entities, attributes, records and relationships model a domain. Conceptual models clarify meaning before database implementation. Consistent identifiers reduce ambiguity.

Deeper understanding

To apply data organisation, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• entity: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• attribute: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• record: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• relationship: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

Students and courses form entities linked by enrolments.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of data organisation in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Relational databases

Core explanation

Tables contain rows and columns; primary keys identify rows and foreign keys link tables. Normalisation reduces redundancy, though performance needs may justify trade-offs.

Deeper understanding

To apply relational databases, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• DBMS: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• primary key: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• foreign key: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• normalisation: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

Orders refer to customers through a customer identifier.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of relational databases in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Queries and transactions

Core explanation

SQL retrieves and modifies data. Transactions should preserve consistency and handle concurrent activity; operational systems prioritise correct, fast updates.

Deeper understanding

To apply queries and transactions, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• SQL: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• transaction: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• ACID: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• OLTP: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A transfer updates two balances as one transaction.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of queries and transactions in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Data warehousing and analytics

Core explanation

Warehouses integrate historical data for analysis. ETL/ELT pipelines collect, clean and transform data; analytical models support reporting and trend exploration.

Deeper understanding

To apply data warehousing and analytics, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• data warehouse: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• ETL: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• OLAP: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A retailer compares three years of sales by region.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of data warehousing and analytics in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Big data and quality

Core explanation

Volume, velocity and variety describe common data challenges. Accuracy, completeness, consistency and timeliness determine fitness for use. More data does not guarantee better decisions.

Deeper understanding

To apply big data and quality, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• big data: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• data quality: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• metadata: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

Duplicate customer records distort campaign results.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of big data and quality in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Knowledge management

Core explanation

Tacit knowledge resides in experience; explicit knowledge is documented. Organisations capture, share and reuse lessons while respecting access rights and context.

Deeper understanding

To apply knowledge management, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• tacit knowledge: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• explicit knowledge: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• knowledge management: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A support team builds a searchable troubleshooting knowledge base.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of knowledge management in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

CHAPTER 7 | E-Business and E-Commerce

CHAPTER ROADMAP

Digital commerce joins customer experience, transactions, operations and trust across connected systems.

Learning outcomes

1. Explain and apply: Scope and models.
2. Explain and apply: Customer journey.
3. Explain and apply: Digital payments and trust.
4. Explain and apply: Operations integration.
5. Explain and apply: Platforms and networks.
6. Explain and apply: Metrics and challenges.

Concept map

BUSINESS QUESTION → DATA / PEOPLE / PROCESS → INFORMATION SYSTEM → ACTION → MEASURABLE OUTCOME

How to use this chapter

Read the six topic pages, explain the definitions aloud, study the scenario, and answer the questions without looking. Then check the model answers. This guide is original educational commentary, not a reproduction of the textbook.

Big picture

Digital commerce joins customer experience, transactions, operations and trust across connected systems. The central skill is to connect terminology to a real organisational decision. Whenever you meet a new concept, identify its purpose, required inputs, expected benefits, limitations, and the people responsible for it.

Scope and models

Core explanation

E-business covers digital business processes; e-commerce focuses on buying and selling. B2B, B2C and C2C identify participants, while platform models connect multiple groups.

Deeper understanding

To apply scope and models, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• e-business: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• e-commerce: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• B2B: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• B2C: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• C2C: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A marketplace connects independent sellers with consumers.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of scope and models in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Customer journey

Core explanation

Discovery, evaluation, checkout, fulfilment and support form a connected journey. Friction, accessibility and transparency influence conversion and retention.

Deeper understanding

To apply customer journey, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• customer journey: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• conversion: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• cart abandonment: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A confusing mobile checkout causes customers to leave.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of customer journey in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Digital payments and trust

Core explanation

Payment gateways, authentication, fraud detection and secure transport support transactions. Merchants must manage chargebacks, privacy and operational failures.

Deeper understanding

To apply digital payments and trust, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• payment gateway: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• fraud detection: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• chargeback: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A merchant requests extra verification for suspicious purchases.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of digital payments and trust in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Operations integration

Core explanation

Order management, inventory, logistics and customer support must exchange timely data. Integration failures can create overselling and late delivery.

Deeper understanding

To apply operations integration, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• order management: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• fulfilment: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• API: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

Stock availability updates after every online purchase.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of operations integration in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Platforms and networks

Core explanation

Network effects can increase platform value as participation grows, but create concentration, governance and lock-in risks. Ratings and moderation build trust.

Deeper understanding

To apply platforms and networks, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• network effect: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• platform: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• disintermediation: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A marketplace grows more useful with additional buyers and sellers.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of platforms and networks in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Metrics and challenges

Core explanation

Measure conversion, average order value, acquisition cost, retention and returns. Evaluate international tax, regulation, accessibility and sustainability issues.

Deeper understanding

To apply metrics and challenges, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• conversion rate: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• CAC: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• retention: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A business improves retention instead of buying ever more advertisements.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of metrics and challenges in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

CHAPTER 9 | Social Computing

CHAPTER ROADMAP

Social technologies create collaborative value but also amplify privacy, reputation and information-quality risks.

Learning outcomes

1. Explain and apply: Social platforms.
2. Explain and apply: User-generated content.
3. Explain and apply: Business applications.
4. Explain and apply: Social commerce.
5. Explain and apply: Harms and misinformation.
6. Explain and apply: Responsible strategy.

Concept map

BUSINESS QUESTION → DATA / PEOPLE / PROCESS → INFORMATION SYSTEM → ACTION → MEASURABLE OUTCOME

How to use this chapter

Read the six topic pages, explain the definitions aloud, study the scenario, and answer the questions without looking. Then check the model answers. This guide is original educational commentary, not a reproduction of the textbook.

Big picture

Social technologies create collaborative value but also amplify privacy, reputation and information-quality risks. The central skill is to connect terminology to a real organisational decision. Whenever you meet a new concept, identify its purpose, required inputs, expected benefits, limitations, and the people responsible for it.

Social platforms

Core explanation

Profiles, connections, feeds and communities support interaction and content sharing. Algorithms influence visibility and engagement.

Deeper understanding

To apply social platforms, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• social computing: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• social network: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• feed: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A professional community shares advice and job opportunities.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of social platforms in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

User-generated content

Core explanation

Reviews, forums, wikis and collaborative documents distribute knowledge creation. Moderation and verification improve usefulness.

Deeper understanding

To apply user-generated content, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• UGC: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• wiki: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• online community: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

Customers post product reviews that guide purchasing.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of user-generated content in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Business applications

Core explanation

Social listening, support, marketing and recruitment use social interactions. Organisations need consistent responses and appropriate consent.

Deeper understanding

To apply business applications, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• social listening: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• social CRM: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• engagement: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A support team identifies recurring defects in public feedback.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of business applications in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Social commerce

Core explanation

Discovery, recommendations and purchasing increasingly overlap. Influencers and social proof can help customers but raise disclosure and manipulation concerns.

Deeper understanding

To apply social commerce, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• social commerce: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• social proof: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• influencer: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A creator links a product demonstration to a shop.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of social commerce in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Harms and misinformation

Core explanation

Bots, trolls, deepfakes, echo chambers and misleading content distort communication. Verify provenance and avoid sharing unconfirmed claims.

Deeper understanding

To apply harms and misinformation, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• bot: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• deepfake: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• misinformation: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• echo chamber: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A fabricated video damages a company reputation.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of harms and misinformation in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Responsible strategy

Core explanation

Choose meaningful metrics, moderate fairly, protect user data and design for accessibility. Engagement alone is not proof of positive outcomes.

Deeper understanding

To apply responsible strategy, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• moderation: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• sentiment: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• accessibility: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A university measures resolved student questions, not only likes.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of responsible strategy in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

CHAPTER 10 | Information Systems within the Organization

CHAPTER ROADMAP

Operational and managerial systems coordinate business functions through shared information.

Learning outcomes

1. Explain and apply: Operational systems.

2. Explain and apply: Management reporting.

3. Explain and apply: Functional information systems.

4. Explain and apply: ERP systems.

5. Explain and apply: Collaboration and workflows.

6. Explain and apply: Implementation and measurement.

Concept map

BUSINESS QUESTION → DATA / PEOPLE / PROCESS → INFORMATION SYSTEM → ACTION → MEASURABLE OUTCOME

How to use this chapter

Read the six topic pages, explain the definitions aloud, study the scenario, and answer the questions without looking. Then check the model answers. This guide is original educational commentary, not a reproduction of the textbook.

Big picture

Operational and managerial systems coordinate business functions through shared information. The central skill is to connect terminology to a real organisational decision. Whenever you meet a new concept, identify its purpose, required inputs, expected benefits, limitations, and the people responsible for it.

Operational systems

Core explanation

TPS records routine transactions accurately and reliably. Examples include payroll, reservations, stock movements and invoicing.

Deeper understanding

To apply operational systems, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• TPS: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• transaction: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• operational data: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A point-of-sale system updates stock after checkout.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of operational systems in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Management reporting

Core explanation

MIS turns transactions into recurring summaries; DSS supports scenario exploration. Decision quality depends on assumptions and underlying data.

Deeper understanding

To apply management reporting, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• MIS: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• DSS: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• structured decision: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A manager compares monthly sales with budget.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of management reporting in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Functional information systems

Core explanation

Accounting, finance, marketing, HR and operations have distinct information needs. Integration avoids duplicated records and contradictory reports.

Deeper understanding

To apply functional information systems, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• FAIS: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• functional silo: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• integration: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

HR data flows into payroll without manual re-entry.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of functional information systems in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

ERP systems

Core explanation

ERP standardises and integrates processes through common modules and shared data. Benefits require change management, configuration and careful migration.

Deeper understanding

To apply erp systems, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• ERP: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• module: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• master data: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

Odoo connects purchasing, inventory and invoicing.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of erp systems in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Collaboration and workflows

Core explanation

Workflow systems assign tasks, approvals and alerts; collaboration tools enable coordination. Too many notifications can reduce productivity.

Deeper understanding

To apply collaboration and workflows, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• workflow: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• collaboration: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• approval: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

A purchase request automatically reaches the correct approver.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of collaboration and workflows in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Implementation and measurement

Core explanation

Set clear objectives, train users, clean data and monitor adoption. Avoid measuring success only by whether software was installed.

Deeper understanding

To apply implementation and measurement, begin by identifying the practical problem and the stakeholders affected. Separate the intended benefit from the evidence that the approach works. Ask which information must be reliable, what process changes are required, and which failure modes could make the outcome worse. These questions turn a definition into an analytical tool.

Key terms and working definitions

• change management: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• adoption: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

• KPI: A core concept in this topic. Explain it in your own words, describe its organisational purpose, and distinguish it from related terms.

Applied example

An ERP project tracks order accuracy after go-live.

Process diagram

INPUTS → VALIDATE / ORGANISE → APPLY CONCEPT → REVIEW RESULT → IMPROVE

Common mistakes and critical thinking

Avoid assuming that purchasing a tool automatically improves results. Check data quality, organisational fit, user adoption, cost, ethics and security. A technically possible solution may be economically or socially inappropriate.

Revision questions

Q1. What is the purpose of implementation and measurement in an organisation?

Q2. How could you evaluate the example above?

Q3. Identify one risk or limitation and propose a safeguard.

Model answers

A1. It addresses the problem described in the core explanation by improving how information, decisions or activities are organised.

A2. Define a measurable outcome before implementation; compare results before and after while checking user experience and data reliability.

A3. Risks include weak data, privacy exposure, poor adoption or incorrect assumptions. Match the safeguard to the risk and assign an accountable owner.

Sources and scope

Rainer, R. Kelly and Brad Prince (2022), Introduction to Information Systems, 9th ed., International Adaptation. Chapter titles verified against the Wiley publisher table of contents. Original study notes; supplement with the UOC assigned readings and official chapter case studies.